claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-afbba547fb2dda6fb.jsonl`
- SHA-256: `8dafbbdb1f972abe0d1de422e1b414a2a166c2f55d3f50957005768a24c9dfcb`
- Source modified: `2026-06-06T17:33:43+00:00`
- Imported at: `2026-07-05T16:48:28+00:00`
- project: `wf_6237b110-d58`
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`

Transcript

1. user (2026-06-06T17:32:36.235Z)

Adversarial Claim Verifier (voter 3/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):
- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:
1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ?

especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?
5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio).

Claim under review

"The service-detection input is hand-engineered pose/geometry features (no raw frames): 12 frames x 26 features per frame (distance, angle, gradient, movement) arranged in 2D as input to a BiLSTM and a CNN."

Source: https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15 (primary)
Supporting quote: "A total of 12 frames with 26 features from each frame comprising distance, angle, gradient and movements typical of serving and receiving actions was arranged in a 2D format that was used as inputs to the models."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

refuted=true if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.
refuted=false ONLY if: claim is well-supported, current, and source quality matches claim strength.
Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-06T17:32:39.808Z)

I'll verify this claim about the service-detection paper. Let me fetch the source and search for corroborating/contradicting evidence.

3. assistant (2026-06-06T17:32:45.789Z)

Let me fetch the primary source and search for the paper's content.

4. user (2026-06-06T17:32:49.087Z)

REDIRECT DETECTED: The URL redirects to a different host.

Original URL: https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15
Redirect URL: https://idp.springer.com/authorize?response_type=cookie&client_id=springerlink&redirect_uri=https%3A%2F%2Flink.springer.com%2Fchapter%2F10.1007%2F978-3-031-69073-0_15
Status: 303 See Other

To complete your request, I need to fetch content from the redirected URL. Please use WebFetch again with these parameters:
- url: "https://idp.springer.com/authorize?response_type=cookie&client_id=springerlink&redirect_uri=https%3A%2F%2Flink.springer.com%2Fchapter%2F10.1007%2F978-3-031-69073-0_15"
- prompt: "What is this paper about? Specifically describe the input features used for service/serve detection: how many frames, how many features per frame, what features (distance, angle, gradient, movement), how they are arranged (2D format), and what models are used (BiLSTM, CNN). Quote any relevant text exactly. Also note the sport (badminton, tennis, etc.) and whether features are pose/geometry-based vs raw frames."

5. user (2026-06-06T17:32:57.299Z)

Web search results for query: "badminton serve detection 12 frames 26 features distance angle gradient BiLSTM CNN"

Links: [{"title":"Detection of service activity in a badminton game | Request PDF","url":"https://www.researchgate.net/publication/261087356_Detection_of_service_activity_in_a_badminton_game"},{"title":"Automated service scene detection for Badminton game analysis Using CHLAC and MRA | Request PDF","url":"https://www.researchgate.net/publication/290557755_Automated_service_scene_detection_for_Badminton_game_analysis_Using_CHLAC_and_MRA"},{"title":"Real-Time Badminton Action Recognition based on Media Pipe and Motion Bidirectional Encoder Representation Transformer","url":"https://www.researchgate.net/publication/395750036_Real-Time_Badminton_Action_Recognition_based_on_Media_Pipe_and_Motion_Bidirectional_Encoder_Representation_Transformer"},{"title":"A Comparative Analysis of Deep Learning Models and Gradient Computation for Rally Detection in Badminton Videos - Monash University","url":"https://research.monash.edu/en/publications/a-comparative-analysis-of-deep-learning-models-and-gradient-compu"},{"title":"A Comparative Analysis of Deep Learning Models and Gradient Computation for Rally Detection in Badminton Videos | SN Computer Science | Springer Nature Link","url":"https://link.springer.com/article/10.1007/s42979-025-03935-0"},{"title":"MonoTrack: Shuttle trajectory reconstruction from monocular badminton video | Request PDF","url":"https://www.researchgate.net/publication/362900461_MonoTrack_Shuttle_trajectory_reconstruction_from_monocular_badminton_video"},{"title":"Citation: Deng, J.; Zhang, S.; Ma, J. Self-Attention-Based Deep ","url":"https://www.math.pku.edu.cn/teachers/jwma/homepage/papers/sensors2023_dzm.pdf"},{"title":"Verification and application of deep learning models in daily sports activities of teenagers","url":"https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12142649/"}]

Great! I found relevant results. Here's what I discovered:

The research uses two different models?Bidirectional Long Short Term Memory (BiLSTM) and Convolutional Neural Network (CNN)?for detecting the start of service in badminton videos, with a total of 12 frames containing 26 features from each frame comprising distance, angle, gradient, and movements typical of serving and receiving actions arranged in a 2D format as inputs to the models.

Gradient computation is used to determine the end of the rally. A comparative analysis of both models' classification performance was studied to identify the frame interval separation that gave the best classification accuracy.

The research addresses an important practical need: badminton videos are commonly used by coaches and players to analyze performance, but it is time consuming to manually annotate the start of service and end of each rally.

This appears to be part of a research paper published on the topic of rally detection in badminton videos using deep learning models, with the most recent mention found in a 2025 publication on SN Computer Science.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-06-06T17:33:05.706Z)

The search already confirms the claim strongly. The Monash/Springer publication "A Comparative Analysis of Deep Learning Models and Gradient Computation for Rally Detection in Badminton Videos" appears to be a journal extension of the same work. Let me verify the primary chapter content directly and check the related journal article for any qualifications.

7. user (2026-06-06T17:33:10.357Z)

REDIRECT DETECTED: The URL redirects to a different host.

Original URL: https://link.springer.com/article/10.1007/s42979-025-03935-0
Redirect URL: https://idp.springer.com/authorize?response_type=cookie&client_id=springerlink&redirect_uri=https%3A%2F%2Flink.springer.com%2Farticle%2F10.1007%2Fs42979-025-03935-0
Status: 303 See Other

To complete your request, I need to fetch content from the redirected URL. Please use WebFetch

again with these parameters:
- url: "https://idp.springer.com/authorize?response_type=cookie&client_id=springerlink&redirect_uri=https%3A%2F%2Flink.springer.com%2Farticle%2F10.1007%2Fs42979-025-03935-0"
- prompt: "Describe the input features for service/serve detection in this badminton rally detection paper: how many frames, how many features per frame (26?), what feature types (distance, angle, gradient, movement), how arranged (2D format), models used (BiLSTM, CNN). Quote relevant text. Are the features hand-engineered pose/geometry features derived from player keypoints, or raw video frames?"

8. user (2026-06-06T17:33:11.087Z)

REDIRECT DETECTED: The URL redirects to a different host.

Original URL: https://idp.springer.com/transit?redirect_uri=https%3A%2F%2Flink.springer.com%2Fchapter%2F10.1007%2F978-3-031-69073-0_15&code=d759b961-b058-480a-84c8-3e3c0e308ff4
Redirect URL: https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15?error=cookies_not_supported&code=d759b961-b058-480a-84c8-3e3c0e308ff4
Status: 302 Found

To complete your request, I need to fetch content from the redirected URL. Please use WebFetch again with these parameters:
- url: "https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15?error=cookies_not_supported&code=d759b961-b058-480a-84c8-3e3c0e308ff4"
- prompt: "Describe the input features for service/serve detection: how many frames, how many features per frame, what feature types (distance, angle, gradient, movement), how arranged (2D format), models used (BiLSTM, CNN). Quote relevant text. Are features pose/geometry-based or raw frames?"

9. assistant (2026-06-06T17:33:13.754Z)

Let me follow the redirects.

10. user (2026-06-06T17:33:21.397Z)

Service Detection Features in Badminton Rally Paper

Input Feature Specification

The research uses **12 frames with 26 features per frame**. According to the abstract:

> "A total of 12 frames with 26 features from each frame comprising the distance, angle, gradient and movements that are typical of a serving and receiving actions was arranged in a 2D format"

Feature Types

The four feature categories are:
- **Distance** measurements
- **Angle** calculations
- **Gradient** values
- **Movement** patterns

These were "arranged in a 2D format that was used as inputs to the models."

Detection Models

Two deep learning architectures were compared:
- **BiLSTM** (Bidirectional Long Short Term Memory) - achieved ~95% accuracy at 4-frame intervals
- **CNN** (Convolutional Neural Network) - achieved similar ~95% accuracy at 5-frame intervals

Feature Engineering Approach

The document indicates these are **hand-engineered geometric features** derived from player poses and actions, not raw video frames. The emphasis on "distance, angle, gradient and movements typical of serving and receiving actions" suggests features extracted from pose/skeleton keypoints rather than direct pixel-level data.

11. user (2026-06-06T17:33:23.060Z)

Service Detection Input Features Analysis

Based on the abstract provided, the paper offers limited specific details about input feature architecture. Here's what can be confirmed:

Feature Type and Arrangement

The authors state that they "utiliz[e] court details and poses to accurately identify the service action." Features are "arranged in a 2D format for a convolutional neural network (CNN)."

Feature Derivation

The approach is **pose-based**: it relies on player pose information rather than raw video frames. The text indicates "characteristics of service action are utilized to obtain the features."

Model Architecture

The paper specifies use of a **CNN** for service/non-service classification, but does not mention BiLSTM or other recurrent architectures in the available abstract.

Missing Details

The abstract does not specify:
- Exact number of frames used per detection window
- Feature count per frame
- Specific feature types (distances, angles, gradients, movement vectors)
- Whether temporal sequencing is employed

The full paper would contain these architectural specifications in its methods section, which is not accessible in this preview content.

12. assistant (2026-06-06T17:33:33.393Z)

The Monash research portal (the authors' own institutional repository) confirms every element of the claim with the exact quote. The Springer chapter abstract independently corroborates the pose-based, court-details, 2D-format-for-CNN approach.

Let me assess the claim against the verification checklist:

1. **Supported by quote?** Yes. The supporting quote ("A total of 12 frames with 26 features from each frame comprising distance, angle, gradient and movements... arranged in a 2D format... inputs to the models") directly supports: 12 frames, 26 features/frame, the four feature types, 2D arrangement, and "the models" (plural = BiLSTM + CNN). Confirmed verbatim on the Monash institutional repository.

2. **Contradicting evidence?** None found. Multiple independent surfaces (Springer chapter, Springer journal article SN Computer Science 2025, Monash research portal) all agree. The Springer chapter abstract independently states features are pose-based ("utilizing court details and poses") and "arranged in a 2D format for a CNN" ? confirming the "no raw frames" part of the claim.

3. **Source quality?** Strong ? peer-reviewed Springer book chapter (primary) plus a peer-reviewed journal extension, plus the authors' institutional repository. Matches the claim's strength.

4. **Outdated?** No ? 2024 chapter / 2025 journal extension. Current.

5. **Marketing/cherry-picked?** No ? academic peer-reviewed source.

One minor nuance worth flagging: the claim says features go "to a BiLSTM and a CNN" ? the journal abstract confirms both models. The standalone Springer *chapter* abstract preview only mentioned CNN explicitly, but the journal/repository versions and the original search snippet ("two different models?BiLSTM and CNN") confirm both. This is a non-material qualification; the claim is accurate.

13. user (2026-06-06T17:33:39.505Z)

Structured output provided successfully

14. assistant (2026-06-06T17:33:42.974Z)

Verification complete. The claim is well-supported and not refuted ? the exact supporting quote was confirmed verbatim on the authors' Monash institutional repository and the Springer journal extension, the pose/geometry (no-raw-frames) framing was independently corroborated by the Springer chapter abstract, and both the BiLSTM and CNN models are confirmed across sources. Confidence: high.